Achernar

Data Collection & Ingestion Plan

What has to happen outside, what has to happen in the code,
and which of the nine models each night of data actually feeds

This document is the bridge between *achernar_v4.pdf* §7.6–§7.7 (which says what the models are) and the repository as it exists on August 7, 2026 (which says what can actually be captured). It is organised so that Part I tells you what is broken, Part II tells you where bytes go, Part III tells you what each model needs, and Part IV tells you exactly what to do at the eyepiece.
Nothing here assumes a model can be trained on data the current code cannot record. Where the code cannot record something, that is stated as a blocking defect with a named file, not as a wish.

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Part I Instrument and software readiness

1 Purpose, scope, and how to read this

1.1 The question this document answers

Every model in §7.6 of the specification is currently a stub, a reference graph with random weights, or a classical estimator with no fitted constants. The reason is not that the architectures are unwritten — most are — but that none of them has ever seen a photon collected by this system. The gap between "the code compiles" and "the model ships" is entirely a data gap.

This document specifies that data: what physically has to be photographed, with which instrument, under what conditions, how many times, what has to be written down at the time because it cannot be recovered afterwards, and what the software has to be able to do before any of it is worth collecting.

1.2 Reading order

- **Part I** audits the daemon's real capture and sensor-control capability, inventories the instruments available, and computes what each optical combination can physically resolve. It ends with a list of defects that must close before the first useful night.
- **Part II** traces the ingestion path that already exists in the repository — `capture_attempts` → `frames` → `frame_assets` → `frame_measurements` — and identifies where it does not yet reach.
- **Part III** takes each of the nine models in turn and states its data contract: label source, required covariates, grouping key, sample budget, which rig can produce it, and what blocks it today.
- **Part IV** is the field material. It is written to the same level of detail as the specification chapters: numbered procedures, exact commands, frame counts, abort criteria, and the log schema.

1.3 Conventions and standing rules

Rule 0 — Citation convention. A reference written "spec §7.6.4" points at *achernar_v4.pdf*. A bare "§7.4" points at a section of *this* document. Both documents happen to have a §7, so the word *spec* is load-bearing wherever it appears.

Rule 1 — Manual first. Every procedure in Part IV is executable by hand. Automation is an overlay, never a prerequisite. If the daemon is down you should still be able to complete the night and ingest afterwards.
Rule 2 — Unknown is a value. If a quantity was not measured, it is recorded as absent, never as a plausible default. The schema already models this (`frames.temperature_state` ∈ {`measured`, `stale`, `missing`}); the field log must too. A guessed ambient temperature will silently corrupt the Focus Drift Forecaster and there is no way to detect it later.
Rule 3 — Raw stays raw. Nothing that is written to disk during a session is calibrated, stretched, debayered, or scaled. Calibration constants travel in the FITS header and the device profile. This is already how `ols_decode_raw` behaves and the field procedures must not undo it.
Rule 4 — The group is the run, not the frame. Splits are by focus run and by night, never by frame or by star. Two stars from the same frame are near-duplicates; a random split will produce a model that looks excellent and generalises to nothing.

2 Capability audit: what the daemon can capture today

This section was written by reading the tree, not the previous handoff notes. File and line references are to the working tree as of August 7, 2026.

2.1 The capture path, end to end

There are two distinct paths through the camera layer, and they have different maturity.

Path A — streaming / live view (`POST /camera/stream/start`)

`OlsCamera::start_stream` (`hardware/camera.rs`) opens a format, pushes frames into a store, and the vision loop (`vision/vision_loop.rs`) converts each to a `LumaFrame`, estimates background, detects stars, and computes classical HFR in `vision/quality.rs`. This path is the one the autofocus routine uses. It is the most complete part of the system: 1254 lines of star-detection and quality code with an explicit saturation gate, an explicit "gate not performed" state, and no fabricated scalars.

Path B — still capture (`POST /camera/capture`)

`OlsCamera::capture_still` (`hardware/camera.rs`, § line 671) drives a DSLR through gphoto2: pause live view, apply and *read back* shutter and ISO, timestamp, trigger once, drain events to `CAPTURE_COMPLETE`, download every companion asset, resume. The returned `StillCapture` carries every asset the body produced (native RAW, camera JPEG, and any sidecar), and `storage/capture_store.rs` commits them with a stage-hash-manifest-fsync- transaction-rename protocol and a startup `reconcile()`.

RAW decoding goes through `ols_decode_raw` into `hardware/raw_decode.rs`, which produces **raw sensor ADU** — no black subtraction, no white balance, no rescaling — and `fits.rs` writes the calibration constants into the header (`BUNIT`, `BLKLEVEL`, `CBLACK0..3`, `SATURATE`, `XBAYROFF`, `YBAYROFF`, `BAYERPAT`).

> **Path B has never been compiled.** The RAW decode work and the `capture_still` implementation were written in an environment with no Rust toolchain, no LibRaw and no OpenCV. `cargo build` on the Mac is the first real test of roughly 1,500 lines. Budget a full session for this before any observing is planned around it. See BP-0 in Part IV.

2.2 Driver matrix

`hardware/capabilities.rs` reports, at runtime, which drivers were actually compiled into the bridge. The three-way distinction it makes — *compiled in*, *compiled out*, *exposed in Rust but never wired to the bridge* — is exactly the distinction that matters for planning, so it is reproduced here.

Subsystem	Driver	Status in code	Relevant to us?
Camera	`sim`	always available	yes — pipeline smoke tests
Camera	`alpaca`	always available	only via a bridge host
Camera	`gphoto2`	compiled if `libgphoto2` found	**yes — all 5 DSLR/mirrorless**
Camera	`uvc`	compiled if `libuvc` found	**yes — SV105**
Camera	`asi`	compiled if ASI SDK present	only if a loaner has one
Camera	`svb`	compiled if SVBony SDK present	*not* the SV105 (see §3.1)
Camera	`toup`	compiled if ToupTek SDK present	unlikely
Camera	`indi`	compiled if `libindi` found	fallback for loaner gear
Camera	`indigo`	compiled if INDIGO found	unlikely
Camera	`wdir`	**Linux only**	see §2.4 defect D3
Mount	`alpaca`	always available	loaner GoTo, via bridge
Mount	`sim`	always available	testing
Mount	`indi`	compiled if `libindi` found	loaner GoTo, native
Mount	serial/LX200	`hardware/serial_mount.rs` is a 12-line TODO	**absent**
Focuser	`alpaca`	always available	only via a bridge host
Focuser	`sim`	always available	testing
Focuser	`indi`	compiled if `libindi` found	**the realistic option**
Focuser	`moonlite`	reported `disabled` — Rust only, no bridge	absent
Focuser	USB serial	reported `disabled` — Rust only, no bridge	absent
Filter wheel	any	`hardware/filter_wheel.rs` is `todo!()`	**absent**
Env. sensors	any	`hardware/sensors.rs` is `todo!()`	**absent**

2.3 What exists above the hardware layer

It is worth being precise about how much is *not* the problem, because it changes where effort should go.

Area	State
Star detection / HFR	Complete and careful. `vision/quality.rs`, 1254 lines. Explicit rejection accounting (saturated, edge, blended), an explicit `SaturationGate` state distinguishing "no saturated stars" from "never checked".
Background estimation	`vision/background.rs` — implemented, saturation-aware.
Trailing (classical)	`vision/trailing.rs` — 223 lines, implemented.
FITS writer	`fits.rs` — 507 lines, writes sensor-ADU Bayer frames with full calibration headers.
Capture durability	`storage/capture_store.rs` — staging, hashing, manifest, fsync, transaction, atomic rename, startup reconcile.
Schema	5 migrations, single authority. `frame_assets` models lineage; `frame_measurements` is append-only and versioned by pipeline/-model/preprocessing.
ONNX serving	`vision/fqe_runner.rs` loads a hash-bound model bundle and publishes shadow-only fields. `models/onnx_runner.rs` generic path is still `todo!()`.
Debayer / stacking / calibration application	4-line TODO files. Not needed for training data.
Plate solving	`processing/plate_solver.rs` — 40 lines. **Needed** (see §7.9).

2.4 Defects that block data collection

These are ordered by how much data they spoil, not by how hard they are.

D1 — Sensor saturation point is not declared, and the ML side reads the wrong field

The daemon models this correctly. `vision/quality.rs` distinguishes `max_adu` (the container ceiling: 255 for an 8-bit stream, 65535 for a 16-bit one) from `saturation_adu` (the sensor's linear saturation point, `Option<f32>`), and `resolve_saturation_adu` deliberately returns `None` for a 16-bit container rather than 65535, so that the gate reports "not performed" instead of silently passing every star.

Two things are still wrong:

1. **The value is never set.** `[camera] saturation_adu` is commented out in `achernar.example.toml`. Until it is measured per body and written in, every DSLR frame you take will carry `saturation_gate = not_performed`.
2. **The dataset builder reads `max_adu`, not `saturation_adu`.** `ml/data_pipeline/build_focus_quality_dataset.py` computes its saturation level as `max_adu * 0.98` with a default of 64000. On a 14-bit Canon delivered in a 16-bit container, real saturation is at ∼16383 ADU; the builder will use 64230 and admit every clipped star into the training set. Clipped stars have artificially flat cores and therefore artificially *small* measured HFR, so this biases the brightest, highest-weight samples toward "sharper than truth".

Blocking. Bench procedure BP-2 measures the per-body saturation point and read noise. The one-line change in the dataset builder — prefer `saturation_adu`, and drop frames where it is absent rather than defaulting — must land before the first training run. This is the single highest-value half-hour of code in the document.

D2 — `saturation_adu` is plumbed into streaming but not into still capture

`api/camera.rs` passes `state.camera_config.saturation_adu` into `start_stream` (line 159). No equivalent call appears on the still-capture path. That means Path B frames — the DSLR RAWs, which are precisely the frames that need it — may reach the quality code without it even once D1 is fixed. Verify during BP-0 and wire it through.

D3 — No focuser with readable absolute position

The Zhumell Z114 has a manual rack-and-pinion focuser. There is no encoder, no motor, and no position readout. Consequently:

- The **Focus Offset Predictor** (§7.6.2) cannot be trained — its label *is* a signed step count.
- The **Focus Drift Forecaster** (§7.6.3) cannot be fitted — α is $\Delta p/\Delta T$ in steps per °C.
- The **Focus Quality Estimator** can still be trained (it is deliberately pixels-only and takes no position input), but the *provenance* of each run is degraded: without positions you cannot order samples within a run or verify the sweep was monotonic.

Two remedies, in increasing cost, are given in §3.5 and RB-2. The cheap one — a dial indicator clamped to the drawtube — turns the manual focuser into a 10 µm absolute encoder for about $15 and makes FOP data collectable by hand.

D4 — No filter wheel, and the hardware layer for one is `todo!()`

`hardware/filter_wheel.rs` is two `todo!()` calls. The control layer above it (`filter_wheel/filter_planner.rs`, `filter_bandit.rs`, `filter_snr_scorer.rs`,

`filter_safety.rs`) and the schema (`filter_switch_events`) both exist. The **Filter Order Recommender** (§7.6.8) is a 7-class model over L/R/G/B/Ha/OIII/SII; you own none of those filters and no wheel. See §7.8 — the recommendation is to descope, not to buy.

D5 — No environment sensor

`hardware/sensors.rs` is `todo!()`. Temperature currently comes from `read_temperature()`, i.e. whatever the camera reports, with an honest `temperature_state` when it reports nothing. A DSLR's reported temperature is the sensor die, which self-heats during a session and is quantised to 1 °C — useful as a covariate, useless as ambient. FDF needs ambient. See §7.3 and RB-4 for the manual-logging workaround, which is adequate for a first fit.

D6 — Plate solving is a stub

40 lines in `processing/plate_solver.rs`. This blocks: verified pixel scale (BP-4), target identity on every frame (needed for the AstroM3 edge retrofit, §7.9), airmass and moon-separation covariates computed from a *solved* pointing rather than a claimed one, and the `plate_solve_events` table which already exists. The cheapest path is to shell out to a local `solve-field` (astrometry.net) install rather than implement a solver.

D7 — COC class count disagrees between spec and code

§7.6.4 of the specification defines a *binary* classifier: `Linear(1024,1) + Sigmoid`, focal loss with $\alpha = 0.25$, $\gamma = 2.0$, temperature scaling on a held-out set. `ml/models/cloud_classifier.py` declares `num_classes: int = 5` with classes {clear, thin, moderate, heavy, overcast}.

This is not a code detail — it determines what you write on the log card at 2 a.m., and relabelling afterwards is impossible because the sky is gone.

> **Resolution adopted in this document:** collect **5-class** labels in the field, and derive the binary label by thresholding ({clear, thin} → 0, {moderate, heavy, overcast} → 1). Five-class labels can always be collapsed to two; two can never be expanded to five. The model can then be trained either way from the same data. Whichever head ships, the field protocol is unchanged.

3 Instrument inventory and what each rig can actually measure

3.1 Owned

SVBony SV105

Sensor	Sony IMX307, 1920×1080, 2.9 µm pixels, 1/2.8"
Interface	USB 2.0, **UVC class** — appears as a webcam, no vendor SDK
Max exposure	**500 ms**
Output	8-bit video (UVC YUY2/MJPEG); no 16-bit linear readout
Form factor	1.25" barrel, prime focus in the drawtube

Three consequences, all of which shape the plan:

1. **It is a UVC device, not an SVBony-SDK device.** The daemon's `svb` driver (SVBony SDK) will not see it. It must come in through `uvc` (libuvc). On macOS the OS claims UVC devices for AVFoundation, which can prevent libuvc from opening them. **Verify in BP-1**; if it fails, the fallbacks are (a) capture with an AVFoundation-based tool and ingest via a watch folder — but `wdir` is compiled **Linux-only** — or (b) run the daemon in a Linux VM/container with USB passthrough.
2. **8-bit output makes the saturation gate load-bearing.** A star peaks at 255 and clips. `resolve_saturation_adu` correctly returns `Some(max_adu)` for 8-bit containers, so the gate *will* fire — but only if the frame is tagged as an 8-bit stream. Confirm during BP-1.
3. **500 ms ceiling rules out deep sky entirely.** It cannot contribute to the exposure bandit, long-exposure FQE, or sky-background-vs-moon campaigns. It *is* excellent for high-cadence focus sweeps on bright stars, where you want dozens of frames per focuser position in seconds.

Zhumell Z114

114 mm aperture, 450 mm focal length, f/3.95, parabolic primary, tabletop alt-azimuth Dobsonian, 1.25" manual rack-and-pinion focuser, no tracking, no setting circles.

The absence of tracking is an asset, not only a limitation. An untracked exposure trails by exactly $15.041 \cos \delta$ arcseconds per second. That is a *physically computed, exactly known* elongation label, available in unlimited quantity, on any clear night, with no rental. This makes the Z114 the best trailing-detector instrument you have access to — better than a tracked loaner rig, which can only produce negatives. See RB-6.

3.2 Loanable camera bodies (UNCG)

Body	Sensor	Pixels	Pitch	RAW	ADC
Canon T3i (600D)	APS-C 22.3×14.9	5184×3456	4.30 µm	CR2	14-bit
Canon T6S (760D)	APS-C 22.3×14.9	6000×4000	3.72 µm	CR2	14-bit
Canon M50	APS-C 22.3×14.9	6000×4000	3.72 µm	CR3	14-bit
Canon R50	APS-C 22.3×14.8	6000×4000	3.72 µm	CR3	14-bit
Lumix G95	MFT 17.3×13.0	5184×3888	3.34 µm	RW2	12-bit

The ADC column is a guess until BP-2 measures it, and it must not be guessed in the config. The whole point of D1 is that a wrong `saturation_adu` is worse than none. In particular the G95 is listed as 12-bit here because Panasonic MFT bodies commonly are, but several shoot 12-bit in electronic shutter and 12-bit in mechanical too; some are 14. Measure it.

Tethering risk, by body

The daemon drives DSLRs through `libgphoto2`. Support is per-body and per-operation, and the operation that matters is *remote shutter release*, not file download.

Body	Expected gphoto2 status — to be confirmed in BP-1
T3i, T6S	Mature Canon PTP support; remote capture and bulb are the best-established case. **Treat as the primary tethered bodies.**
M50	Listed in the libgphoto2 supported-camera table; users have reported intermittent capture failures on the M50 family. Medium risk.
R50	Not on the official supported list. Reports indicate the camera connects and live view / file listing work, but `-capture-image-and-download` may complete without taking a frame; behaviour differs between macOS and Linux. **High risk — do not plan a rental night around it until BP-1 passes.**
G95	Panasonic tether support in gphoto2 is materially weaker than Canon's. **High risk.** Plan for the fallback: shoot with an intervalometer, ingest RW2 files afterwards as an offline import.

Bodies that cannot be tethered are still useful. A body driven by an intervalometer produces perfectly good frames for COC (§7.4), sky background (§7.6) and trailing (§7.5) — none of which require the daemon to control the shutter. What they *cannot* do is participate in a closed loop: no autofocus sweep, no exposure bandit. Assign untethered bodies to open-loop campaigns and do not waste rental nights on them.

3.3 The loan-programme telescope

The rig arrives shipped, is picked up in Chapel Hill, and is available for a bounded window. That window is the scarcest resource in this entire plan, so it should be spent only on the campaigns that *cannot* be run from the driveway with the Z114. Before booking, get answers to the following — the answers change the plan materially, and two of them can make Part III's Tier 2 disappear entirely.

1. Aperture, focal length, and focal ratio (fixes pixel scale — §3.4).
2. Mount model, and **does it have an ASCOM/Alpaca or INDI driver?** The daemon speaks only Alpaca and INDI for mounts. A hand-controller-only mount is manual-tracking-only from the daemon's point of view.
3. **Is there an electronic focuser, and what protocol?** If yes — and it is INDI- or Alpaca-drivable — then FOP and FDF become collectable during the rental and you do not need to buy a focuser. This is the single most consequential question on the list.
4. Is there a filter wheel or filter drawer, and which filters?
5. Is a guide camera / guide scope included? (Guiding RMS is a context feature for the exposure bandit, §7.7.)

6. Loan duration, shipping lead time, and whether the window can be chosen to fall on a new moon.
7. Power requirement — mains, or 12 V battery? This constrains site choice more than people expect.

3.4 Sampling: what each combination can resolve

Plate scale in arcseconds per pixel is $s = 206.265\,p\,/\,f$ with p in µm and f in mm. Typical Piedmont North Carolina seeing is 2.5–4″; the table uses 3″.

Optic (f)	Camera	"/px	FOV (arcmin)	FWHM @ 3" (px)
Z114 native (450 mm)	SV105	1.33	42.5 × 23.9	2.26
	T3i	1.97	170.4 × 113.6	1.52
	T6S / M50 / R50	1.70	170.4 × 113.6	1.76
	G95	1.53	132.2 × 99.1	1.96
Z114 + 2× Barlow (900 mm)	SV105	0.66	21.3 × 12.0	4.51
	T3i	0.99	85.2 × 56.8	3.04
	T6S / M50 / R50	0.85	85.2 × 56.8	3.52
	G95	0.76	66.1 × 49.6	3.92
80 mm f/6 refractor (480 mm)	SV105	1.25	39.9 × 22.4	2.41
	T6S / M50 / R50	1.60	159.7 × 106.5	1.88
8" f/5 Newtonian (1000 mm)	SV105	0.60	19.1 × 10.8	5.02
	T3i	0.89	76.7 × 51.1	3.38
	T6S / M50 / R50	0.77	76.7 × 51.1	3.91
	G95	0.69	59.5 × 44.6	4.36
8" f/10 SCT (2032 mm)	SV105	0.29	9.4 × 5.3	10.19
	T6S / M50 / R50	0.38	37.7 × 25.2	7.95
11" f/10 SCT (2800 mm)	SV105	0.21	6.8 × 3.8	14.04
	T6S / M50 / R50	0.27	27.4 × 18.3	10.96

The Z114 at native focal length is undersampled at best focus for every camera you have. Nyquist wants FWHM $\gtrsim$ 2 px; the native combinations sit at 1.5–2.3 px. The consequence is specific and important: *the labels near the bottom of the V-curve are quantisation-limited.* The defocused wings are fine — they are many pixels wide — but the region the Focus Quality Estimator most needs to resolve is exactly the region the optics cannot deliver.
Remedy: a 2× Barlow (roughly $40–60). It moves every combination to 3–4.5 px FWHM, which is properly sampled, and it doubles the defocus excursion per millimetre of focuser travel, which makes the dial- indicator scheme in RB-2 more precise in angular terms. For FQE and TED work on the Z114 the Barlow should be treated as mandatory equipment, not an option.

Unverified: whether a DSLR reaches prime focus on the Z114 at all. Many tabletop Newtonians place the focal plane inside the drawtube, so a camera body with a 44–45 mm flange distance cannot reach it. The SV105, being a 1.25" barrel that inserts deep into the tube, almost certainly will. Test this in daylight before planning anything around it — BP-3.

3.5 Rig definitions used throughout this document

A "rig" is the grouping key that matters for generalisation testing: the `rigs` table exists in migration 002 and `focus_runs.rig_id` references it. Hold at least one rig out of training entirely.

ID	Configuration	What it can produce
R1	Z114 + SV105, native	TED (untracked ladder), fast focus sweeps, planetary/lunar. No deep sky.
R2	Z114 + 2× Barlow + SV105	As R1, properly sampled. **Primary FQE rig without rental.**
R3	Z114 + Barlow + T6S or T3i	Longer exposures, 14-bit, tethered focus sweeps. Untracked, so exposures $\lesssim 1$ s.
R4	DSLR + kit lens, fixed tripod	COC, sky background, all-sky. No telescope needed. **Runs in parallel with everything else.**
R5	Loaner scope + loaner mount + DSLR	Tracked long exposure, exposure bandit, FOP/FDF if an electronic focuser is included.

R4 is free and independent. It uses a body, a lens, a tripod and an intervalometer, needs no telescope, no daemon and no rental, and it is the only rig that collects the Cloud/Obstruction Classifier's data. **It should be running on every single night you are outside, including nights whose primary purpose is something else, and including bad nights.**

Part II The ingestion pipeline

4 The path bytes take, as built

4.1 Table by table

The schema is the single authority (`achernar-daemon/migrations/`); the Rust repositories under `src/storage/` are its clients. A single exposure touches four tables.

Table	Role
`capture_attempts`	Written *before* the camera is commanded. Carries `capture_id`, `requested_json`, and a status of `started|committed|failed`. This is what makes "the mount was slewing and the capture silently did nothing" distinguishable from "no capture was requested". For a data campaign this is the row that lets you count what you *attempted*, not just what succeeded — which is the denominator in every yield calculation in Part IV.
`frames`	One row per exposure. Migration 004 widened it from a thin file record into a capture record: `rig_id`, `camera_name`, `capture_type` (`light|dark|flat|bias`), `capture_status`, `exposure_started_at` / `_completed_at`, `requested_exposure_s`, `requested_gain`, `iso`, `shutter_label`, `focuser_position`, `temperature_source`, `temperature_state`, `software_version`, `driver_name`, `acquisition_json`.
`frame_assets`	The bytes. One row per artefact with an explicit `role`: `native_raw`, `camera_jpeg`, `sensor_fits`, `analysis_luma`, `thumbnail`, `calibrated_fits`. Carries `checksum_sha256`, `byte_length`, `bit_depth`, and `derived_from_asset_id` — so the FITS knows it came from the RAW and the luma knows it came from the FITS. `variant` means a reprocessed asset is a new row, not an overwrite. Paths are *relative* to the capture root, so the archive can be moved to an external drive without rewriting the database.
`frame_measurements`	Append-only, one row per (frame, measurement type). `hfr | fwhm | star_count | fqe | cloud | trailing`, each stamped with `pipeline_version`, `model_version`, `preprocessing_version`, plus `valid` and `rejection_reason`. Measurements are opinions produced by a specific version of a specific pipeline; overwriting them inside `frames` would make it impossible to re-evaluate later without destroying what the previous model said.

4.2 The commit protocol

`storage/capture_store.rs` does the following, in order, and the order is the point:

1. Insert the `capture_attempts` row with status `started`.
2. Stage every downloaded asset to `<name>.partial`, hashing while writing (one pass, not two).
3. Write `manifest.v1.json` beside them.
4. `fsync` the files and the directory.
5. Open a SQLite transaction; insert `frames` + all `frame_assets`.
6. Atomically rename each `.partial` to its final name.
7. Commit the transaction; set the attempt to `committed`.
8. *Only then* delete the originals from the camera's card.

`reconcile()` runs at startup and repairs the two failure modes this leaves: committed rows whose files are missing, and staged files with no committed row.

Field consequence. If the laptop dies mid-download, nothing is lost and nothing is half-recorded — but the frame may still be on the camera card. **Do not format cards in the field.** The post-session checklist (Part IV, §D) has a card-retention step for exactly this reason.

4.3 What the ML side reads

`ml/data_pipeline/build_focus_quality_dataset.py` joins `focus_samples` to `frames` via `focus_samples.frame_id`, loads the 16-bit PNG at `frames.local_path`, detects stars, and labels each crop with a SEP/DAOPHOT-style HFR from `ml/labels/psf_measure.py`. It emits an `.npz` keyed by `run_key` (the split/grouping key), plus `rig_id`, `camera_id`, `filter_name`, `night_id` as deployment strata.

`ml/data_pipeline/split_registry.py` then performs grouped splitting — all stars and all frames of one autofocus run stay on the same side — and writes a JSON registry so a split is reproducible.

> The builder's own docstring names the wall: "this needs a populated `achernar.db` and a `frames/` directory produced by real autofocus runs on real rigs. Until that exists the builder has nothing to read." Everything in Part IV exists to produce exactly that.

4.4 Gaps between the pipeline and the campaign

Gap	Description	Blocks
G1	Saturation level read from `max_adu` not `saturation_adu` (D1).	FQE labels
G2	No offline-import path for frames the daemon did not capture. Untethered bodies (G95, possibly R50) and intervalometer runs produce files with no DB row.	R4 entirely
G3	`wdir` watch-folder driver is Linux-only.	G2's obvious fix
G4	No ingestion for a manually recorded environment log (ambient temperature, sky condition label, wind).	FDF, COC
G5	`capture_type` exists but nothing sets it to `bias|dark|flat`.	calibration library
G6	Plate solve stub, so `frames` has no verified pointing, no true pixel scale, no target identity.	TME bookkeeping, airmass covariates

G2 and G4 are the two that must close first

They share one solution: a **sidecar-import tool**. A small Python or Rust utility that takes (a) a directory of camera files and (b) a session log in the JSON schema of Part IV §C, and writes the corresponding `frames` / `frame_assets` rows using the same hashing and manifest discipline as `capture_store.rs`.

This is worth building before the first night, because without it every frame that the daemon did not personally trigger is scientifically orphaned — and that is the majority of what the plan collects. It should reuse `capture_store.rs` rather than reimplement it, and should mark imported rows with `driver_name = "offline_import"` and an `acquisition_json` recording which log file supplied the metadata, so imported and daemon-captured data can always be separated during analysis.

5 Grouping, leakage, and the things that cannot be recovered later

5.1 The grouping hierarchy

Level	Key	Why it is a group
Star	—	Never a split unit. Stars in one frame share seeing, focus, tracking and sky.
Frame	`frames.id`	Never a split unit. Frames in one run share the optical train and the night.
Run	`focus_runs.id`	**The FQE/FOP split unit.**
Night	`night_id`	**The COC/background/bandit split unit.** Seeing, transparency and sky brightness are night-level.
Rig	`rigs.id`	Hold one out entirely — this is the only test of whether the model learned optics or learned physics.

§7.7.2 of the specification additionally requires a *temporal* split (oldest 70 % train / next 15 % val / newest 15 % test), which the current `split_registry.create_group_splits` does not implement — it shuffles groups randomly with a seed. Both are defensible; random-group is a fair test of *rig* generalisation, temporal is a fair test of *drift*. Report both.

Practical consequence for the campaign: a grouped split with a temporal component needs *groups*, not *frames*. Twenty thousand star crops from three focus runs is three data points. This is the single most common way an astro-ML dataset that looks large turns out to be worthless, and it drives every sample budget in Part III.

5.2 Irrecoverable metadata

The following cannot be reconstructed from the pixels afterwards. If it is not written down at the time, the frames that depend on it are unusable for the model that needs it.

Quantity	Needed by	Source
Sky-condition class (5-level)	COC	**You, at the time**
Ambient temperature	FDF, FQE covariate	Probe or hand-logged thermometer
Relative humidity, dew point	FOR, bandit, dew diagnosis	Probe or phone weather
Focuser absolute position	FOP, FDF	Encoder / dial indicator
Which direction the focuser was last moved	FOP (backlash)	Log, or daemon
Wind (Beaufort estimate)	TED (distinguishes wind from tracking error)	**You, at the time**
Whether tracking was on	TED	Log
Moon altitude and illumination	bandit, FOR, background	Computable from time + site
Transparency / naked-eye limiting magnitude	bandit, FOR	**You, at the time**
Dew on the corrector or mirror	everything — it invalidates the night	**You, at the time**
Anything you touched mid-run	everything	**You, at the time**

The last row is the one that gets forgotten. If you bump the tripod, nudge the focuser to check something, swap an eyepiece, or a car's headlights sweep the field — **write the timestamp**. One un-annotated disturbance in the middle of a focus run turns a clean V-curve into a bimodal one, and the fitter will silently converge on the wrong minimum. There is a dedicated *Disturbances* block on the field card for this.

Part III Per-model data requirements

6 How to read the model sections

Each subsection below follows the same eight-field template.

Field	Meaning
Predicts	The target, in its natural units.
Label source	Where ground truth comes from — and whether a human must supply it.
Unit / group	The row of the dataset, and the key splits are made on.
Covariates	What must accompany every row, and where it comes from.
Rigs	Which of R1–R5 can produce it.
Budget	v0 / v1 / v2 sample targets, in *groups* first and rows second.
Blocked by	Defects from §2.4 that must close first.
Runbook	The Part IV procedure that collects it.

The three budget tiers mean:

- **v0 — plumbing.** Enough to prove the dataset builder, the split registry, the training loop and the ONNX export run end to end on real bytes. Not enough to evaluate anything. Expect the model to be bad; the deliverable is a green pipeline.
- **v1 — shadow-eligible.** Enough to train something that beats the classical baseline on a held-out group split and can run in shadow mode alongside the classical authority. Not promotable.
- **v2 — promotion candidate.** Enough groups, rigs, nights and covariate span that a held-out *rig* evaluation is meaningful.

7 Model-by-model

7.1 Model 1 — Focus Quality Estimator (FQE), spec §7.6.1

Predicts	$\mu = \log(\mathrm{HFR}_{\mathrm{px}})$ and $\log \sigma^2$ per star crop, pooled robustly to a frame value. Note the repo deviates from the spec deliberately: per-star not per-frame, log-HFR in *pixels* not arcseconds, and **pixels only, no focuser position** (position would let the network regress HFR from a rig-specific shortcut instead of measuring PSF width).
Label source	Automatic — SEP/DAOPHOT HFR from `ml/labels/psf_measure.py`. **No human labelling.**
Unit / group	Row = one star crop. Group = `focus_runs.id`.
Covariates	`rig_id`, `camera_id`, `night_id`, `filter_name`, `position`, `star_snr`. Position is recorded as provenance, not fed to the net.
Rigs	R2 (primary, no rental), R3, R5.
Budget	**v0:** 20 runs, 1 rig, 2 nights. **v1:** 60 runs, ≥2 rigs, ≥4 nights. **v2:** 150 runs, ≥3 rigs, ≥8 nights, ≥10 °C ambient span, one rig fully held out.
Blocked by	D1 (saturation), D2, and the Barlow (§3.4).
Runbook	RB-2 (manual focuser), RB-3 (motorised).

Why the budget is expressed in runs

A single 21-position sweep with 150 usable stars per frame yields ∼3,000 crops — but they are one group. The binding constraint on a grouped split is the group count, not the row count.

Thirty runs is the practical floor for a train/val/test split with any power at all; 150 runs across three rigs is what makes a held-out-rig evaluation interpretable.

At 21 positions × 5 frames × ∼8 s per position, one sweep is roughly 15 minutes including settling. Three sweeps per hour, four productive hours, gives **∼10 runs per night** — so v1 is four to six clear nights and v2 is fifteen to twenty. That is the real schedule.

What makes a run usable

- Monotonic on each side of the minimum, with the minimum interior to the sweep — not at an endpoint.
- ≥15 detected stars at every position (fewer at the extremes is expected and acceptable down to 8).
- Saturation gate *performed*, and < 5 % of stars rejected as saturated.
- No disturbance logged during the run.
- Ambient temperature drift within the run < 0.5 °C, otherwise the run is a drift measurement rather than a clean V-curve.

7.2 Model 2 — Focus Offset Predictor (FOP), spec §7.6.2

Predicts	Signed focuser steps from the current position to best focus, plus direction logits and an uncertainty.
Label source	Automatic, but *retrospective*: the label is $p_{\text{best}} - p_i$, and p_{best} is only known once the run has converged. **An aborted run produces no labels at all.**
Unit / group	Row = one focus *sample* within a run. Group = run.
Covariates	The 38-feature vector of §7.6.2: 3 quality, 4 classical, 5 position, 4 temperature, 7 filter one-hot, 6 history, 3 direction, 3 rig params, 3 sky.
Rigs	R5 with an electronic focuser; R3 with the dial-indicator scheme at reduced precision.
Budget	**v0:** 40 runs. **v1:** 200 runs (∼4,000 rows). **v2:** 500 runs (∼10,000 rows), ≥3 rigs.
Blocked by	**D3** (no position readout) — hard blocker. D4 degenerates the 7 filter features to a constant.
Runbook	RB-3, and passive harvesting (below).

FOP needs an order of magnitude more runs than FQE, from the same nights. Each run gives FQE thousands of crops but gives FOP only ∼20 rows, one per sampled position, and a LightGBM over 38 features wants ≳10,000 rows. Dedicated sweep nights will never get there.

The only viable strategy is passive harvesting: instrument the daemon so that *every* autofocus run — during any session, for any purpose — writes FOP rows. Then FOP data accumulates as a free byproduct of every other campaign in this document, and of ordinary use afterwards. The schema for this already exists (`focus_runs`, `focus_samples`, `focus_actions`); the work is making sure nothing runs autofocus without logging.

Feature-group coverage, not just row count

The 38 features are grouped, and three groups will be degenerate in your data unless deliberately varied:

- **Filter (7 one-hot).** Constant-zero without a wheel. Train with the block dropped and document it in `norm_params.json`, or the model will carry seven dead inputs into an ONNX graph that a future filtered rig will silently misuse.

- **Temperature (4).** Needs real thermal span. A single 20-minute session has none. Deliberately observe across a night's cooling curve — see RB-4.
- **History / direction (9).** `prev_delta_1/2/3`, `prev_hfr_delta_1/2/3`, `dir_changed_recently`, `steps_since_flip`, `consec_improvements`. These only vary if runs actually reverse direction. A sweep that marches monotonically from one end to the other never flips. **Include reversal runs**: start inside the V, move the wrong way, detect it, reverse. Roughly one third of runs should be reversal runs or the direction head has nothing to learn.

The specification's balanced sampler (§7.6.2, Listing 12) buckets $|\delta|$ at $[0, 5, 15, 40, 120, \infty)$ steps. Sweeps that only ever start near focus leave the two largest buckets empty and resampling cannot invent them. **Deliberately start some runs far out** — at ±120 steps or more.

7.3 Model 3 — Focus Drift Forecaster (FDF), spec §7.6.3

Predicts	Focus position at a future time, given temperature; and a refocus trigger when $\lvert p_{\text{forecast}} - p_{\text{now}} \rvert > 0.5$ CFZ.
Label source	Automatic from consecutive converged runs: $\alpha_{\text{new}} = (p_k - p_{k-1})/(T_k - T_{k-1})$, EWMA'd at 0.8/0.2 into `focuser_profiles.temp_coefficient_steps_per_c`.
Unit / group	Row = consecutive run pair. Group = night × rig.
Covariates	Ambient temperature (**not** sensor die temperature), timestamp, rig, focuser position.
Rigs	R5, or R3 with dial indicator.
Budget	**v0:** one night, 8 runs, ≥4 °C span. **v1:** 3 nights per rig, ≥6 °C each. **v2:** 3 rigs × 3 nights, spanning both a cooling and a warming night if possible.
Blocked by	**D3** and **D5**.
Runbook	RB-4.

This is not really a learned model, and that changes the plan

FDF v1 is a two-state Kalman filter with hand-set noise parameters ($\sigma_p = 2$ steps, $\sigma_{\dot{p}} = 0.5$ steps/min, $\sigma_z = 3$ steps) plus one fitted scalar per rig, α. It needs *few, well-separated, well-thermometered* measurements rather than many noisy ones. Eight clean autofocus runs spread over a 6 °C night beat two hundred runs at constant temperature.

Note also that α is a property of the *optical tube*, not of the camera — an aluminium tube and a carbon one have different coefficients. A loaner scope's α does not transfer to the Z114. Each tube needs its own night.

Manual ambient logging is adequate for v1. A $10 outdoor thermometer read every 15 minutes and written on the card, ingested through the sidecar importer (§G4), gives α to well within the precision the Kalman filter's $\sigma_z = 3$ steps assumes. Do not let D5 block this — implement `hardware/sensors.rs` later, log by hand now.

7.4 Model 4 — Cloud / Obstruction Classifier (COC), spec §7.6.4

Predicts	Sky condition. Spec says binary + sigmoid + focal loss; repo stub says 5 classes. **Collect 5, derive 2** (see D7).
Label source	**Human, at the time of capture.** This is the only model in the system whose labels a person must supply, and they are irrecoverable.
Unit / group	Row = one sky frame. Group = **night**, and within a night, contiguous 10-minute label blocks.
Covariates	Timestamp, altitude/azimuth of the field, moon altitude and illumination, whether a light dome is in frame, humidity.
Rigs	**R4** — body + wide lens + tripod + intervalometer. No telescope, no daemon, no rental.
Budget	**v0:** 2 nights, ≥500 labelled frames. **v1:** 8 nights, ≥2,000 frames per class for {clear, thin} and ≥800 for each of {moderate, heavy, overcast}. **v2:** 20 nights across ≥3 moon phases and ≥2 sites, with a full held-out night per class.
Blocked by	G2 (offline import), G4 (label ingestion).
Runbook	RB-5.

The counterintuitive instruction: go outside when it is cloudy. Clear nights are over-represented in any naturally collected set, and the spec already anticipates this with focal loss ($\alpha = 0.25$, $\gamma = 2.0$). But focal loss reweights a class that exists; it cannot create one. Broken, marginal, scudding-cumulus nights are the *most* valuable nights for COC and the least valuable for everything else — so they cost nothing to spend this way.
Put R4 on the tripod on nights you would otherwise write off.

Hard negatives to collect deliberately

The spec calls for hard-negative mining. These are the ones this site will actually produce:

- Thin high cirrus that a photograph barely shows but which kills transparency.
- Moon glow near a bright moon — *not* cloud, and easily confused with it.
- The Durham / Chapel Hill light dome low in the sky — a bright gradient that is not cloud.
- Aircraft contrails, both fresh and spread.
- A tree branch, roof line, or power line intruding into frame — this is *obstruction*, which the model is also named for and which is trivially easy to collect and easy to forget.
- Dew or frost on the lens — a whole-frame softening that resembles overcast. Photograph it on purpose once before wiping it off.
- Passing car headlights and a neighbour's porch light.

Labelling discipline

Label on a fixed cadence (every 10 minutes) rather than per frame, and record transitions to the minute. Take a phone photo of the sky at each label as an independent cross-check — when a label looks wrong during cleaning, the phone photo is what adjudicates. Photographs of the sky taken with the same body at a fixed exposure are also directly usable as a second view of the same label.

7.5 Model 5 — Trailing / Elongation Detector (TED), spec §7.6.5

Predicts	Classical: frame ellipticity $e = \tilde{\mu}_{0.2}\{1 - b_i/a_i\}$, trigger at $e > 0.15$ with $N \geq 10$, plus position angle. CNN: only for the ambiguous band $0.10 < e < 0.20$ or $N < 10$.
Label source	**Physically computed.** An untracked exposure trails by $15.041 \cos\delta$ arcsec per second, exactly.
Unit / group	Row = one 32×32 star cutout. Group = (field, exposure step, night).
Covariates	Exposure time, declination, plate scale (from BP-4), tracking on/off, wind estimate, position angle of the trail.
Rigs	**R1/R2 for positives** (Z114 is untracked — every frame is a labelled trail), R5 for negatives.
Budget	**v0:** 1 night, 3 fields, 12 exposure steps. **v1:** 3 nights, 5 fields, 12 steps, 20 repeats — ∼3,600 frames, ≳100k cutouts. **v2:** add ≥2 tracked nights of negatives and ≥1 windy night.
Blocked by	BP-4 (need a solved plate scale to convert arcsec to px).
Runbook	RB-6.

The exposure ladder gives a continuous label axis

At R2 (Z114 + Barlow + SV105, 0.66 "/px), on the celestial equator:

t_{exp} (s)	trail (arcsec)	trail (px)	regime
0.017 (1/60)	0.26	0.4	indistinguishable from round
0.05	0.75	1.1	sub-seeing
0.10	1.50	2.3	marginal — **the ambiguous band**
0.20	3.01	4.6	clearly elongated
0.33	4.96	7.5	
0.50	7.52	11.4	

The rows in bold are the ones the CNN exists for. The classical estimator already handles obvious trails; the model is only invoked at $0.10 < e < 0.20$. **Sample the ladder densely between 0.05 s and 0.25 s** — that band is where the training signal lives, and a linear sweep of exposure times will under-sample it.
Declination is the second lever: $\cos\delta$ at $\delta = 70°$ is 0.34, so a high-declination field at 0.30 s trails the same as an equatorial field at 0.10 s. Shooting three declinations × a dense exposure ladder gives a well-covered 2-D label grid from a manual telescope in one evening.

Negatives and confounders

Perfectly round stars are as necessary as trailed ones, and the Z114 cannot make them. Sources: (a) the loaner tracked rig, (b) SV105 exposures short enough that the trail is well below a pixel, at ≤0.02 s. Also collect the confounders that produce elongation *other* than tracking error — coma at the field edge of an f/3.95 Newtonian (which is severe and always present), astigmatism from a pinched mirror, and wind-shake, which produces a *random-walk* smear rather than a linear one. The position-angle output is what distinguishes them: tracking error has a consistent PA across the frame, coma is radial from the optical axis, wind is incoherent.

7.6 Model 6 — Background / SNR Estimator, spec §7.6.6

Predicts	`background_adu`, `background_sigma`, `snr_proxy_median`, `saturation_fraction`. Classical (SEP), always active. Optional DnCNN-lite noise map is v2, GPU-only.
Label source	**No labels — it needs *constants***: `READNOISE` and `SENSOR_SATURATION` appear literally in the spec's Listing 15.
Unit / group	N/A for the classical path.
Covariates	Per (body, ISO, temperature) — gain in e^-/ADU, read noise in e^-, black level, saturation point.
Rigs	**Bench. Indoors. No sky required.**
Budget	5 bodies × 4 ISOs × (20 bias + 12 flat pairs) ≈ 1,400 frames. One evening per body.
Blocked by	Nothing. **This can be done tonight, in any weather.**
Runbook	BP-2.

This is the highest-value work available right now. It requires no clear sky, no rental, no focuser, and no working daemon. It produces the constants that D1 needs, which unblocks the FQE labels, which unblocks the entire focus stack. Every other model in this document consumes its output. If there is one thing to do before reading further, it is BP-2.

The photon-transfer method: for a pair of flats at the same illumination,

$$g = \frac{(\bar{S}_1 + \bar{S}_2) - (\bar{B}_1 + \bar{B}_2)}{\sigma^2_{(S_1 - S_2)} - \sigma^2_{(B_1 - B_2)}} \quad [e^-/\text{ADU}], \qquad \sigma_{\text{read}} = g\,\sigma_{\text{bias}} \quad [e^-]$$

The saturation point is the ADU at which the variance-versus-signal curve turns over — *not* the maximum value in the file, which is the container ceiling. Both go into the device profile; the saturation value additionally goes into `[camera] saturation_adu`.

7.7 Model 7 — Exposure / Gain Recommender (LinUCB), spec §7.6.7

Predicts	An arm from $\{10, 30, 60, 90, 120, 180, 300\}$ s given a 17-dim context.
Label source	The reward, computed after the fact from five components: sub accepted (+1.0), SNR in range (+0.5), sub rejected (−0.5), saturation > 5% (−0.3), trailing (−0.2).
Unit / group	Row = one (context, arm, reward) tuple. Group = night.
Covariates	Sky background, moon illumination, moon–target separation, filter one-hot, target type one-hot, seeing, airmass, guiding RMS.
Rigs	**R5 only.** 300-second exposures require tracking. The SV105 (500 ms ceiling) cannot participate at all.
Budget	**v0:** 200 tuples, uniformly randomised. **v1:** 1,500 tuples, ≥6 nights, ≥3 moon phases. **v2:** 5,000 tuples with ≥2 target types per session.
Blocked by	Rental window; D6 (airmass/moon separation want a solved pointing); no guider means one context feature is permanently absent.
Runbook	RB-8.

Log the propensity or the data is worthless for offline evaluation. A contextual bandit is trained online, so it appears to need no dataset. But you cannot afford to let it explore on live rental nights, which means you need *offline* policy evaluation — and every offline estimator (IPS, doubly robust, self-normalised IPS) requires knowing the probability with which the logging policy chose each action.
Concretely: run a deliberately randomised logging policy. The first 200 subs use a uniform policy ($\pi = 1/7$ for every arm); thereafter ε-greedy with $\varepsilon = 0.2$ recorded per decision. Write the sampled arm *and* its probability into `focus_predictions`-style provenance. Without that column, six nights of rental time produce a dataset on which no unbiased estimate of any alternative policy can be computed.

Store reward components, not the scalar

The five reward terms have different measurement provenance — "sub accepted" is a policy decision, "saturation > 5%" is a measurement, "trailing" is another model's output. Storing only the sum makes it impossible to re-weight the reward later without re-observing. Store all five in `frame_measurements.payload_json` and compute the scalar at training time.

7.8 Model 8 — Filter Order Recommender (FOR), spec §7.6.8

Predicts	Next filter, 7-class over L/R/G/B/Ha/OIII/SII, LightGBM multiclass with SHAP explanations.
Label source	The filter that was actually chosen, plus its realised SNR.
Rigs	**None available.**
Blocked by	**D4** — no filter wheel, no filters, and `hardware/filter_wheel.rs` is `todo!()`.
Runbook	RB-9, documented but descoped.

Recommendation: descope FOR. Do not buy hardware for it.
The reasoning is not just cost. FOR sits at the top of the dependency stack — its feature set consumes "historical rejection rate per filter", "mean HFR per filter", "sky background per filter", "seeing estimate" and "transparency score", every one of which is an output of a model that does not exist yet. Building FOR before FQE, background and COC work means training a recommender on features that are currently constants.
If a loaner rig arrives *with* a wheel, log every filter change through `filter_switch_events` as free data and revisit later. If you specifically want a cheap partial path, a manual 1.25" filter drawer with L + UHC + Ha is around $150 and yields a 3-class variant — but that is a different model from the one in §7.6.8 and should be documented as such, not silently substituted.

7.9 Model 9 — Tri-Modal Embedding (TME / AstroM3), spec §7.6.9

Predicts	A shared embedding space; used for retrieval, target ranking, anomaly detection.
Deployments	**(a) Cloud/planning:** AstroM3 as published — light curve + spectrum + metadata. **Weights already trained and available** (`scripts/download_astrom3.sh` → `ml/checkpoints/astrom3/`). **(b) Edge/-daemon:** `AstroM3Edge` — a retrofit with a *new image encoder* over image crop + capture metadata + on-mount telemetry, reusing AstroM3's Informer, MetaModel and CLIPLoss.
Field data needed	**Deployment (a): none.** It consumes archival survey data, not your photons. **Deployment (b): the new image encoder is the only untrained part**, and contrastive training from scratch is out of reach at your data scale — AstroM3 was trained on $\sim 10^5$ objects.
Rigs	Any — as a byproduct.
Blocked by	**D6** (plate solving).

TME needs no dedicated observing at all — it needs bookkeeping.
Two concrete asks, both cheap:

1. **Every frame must carry a resolved target identity.** That means plate solving works (D6) and the solved centre is matched to a catalogue object. This is free once `solve-field` is wired in, and it is independently valuable for the bandit's airmass and moon-separation features.
2. **Train the edge image encoder by distillation, not contrastively from scratch.** Freeze AstroM3's metadata encoder and its embedding space; train only a small image adapter to map your crops into that existing space, supervised by the frozen encoders' output for the same resolved object. This needs thousands of frames, not 10^5 objects, and every other campaign in this document produces them for free.

Set the threshold at **≥10,000 plate-solved frames with resolved target identities** before attempting the adapter. You will cross it during the FQE and TED campaigns without shooting a single dedicated frame.

8 Priority order and campaign schedule

8.1 Dependency tiers

Tier	Precondition	Yields
T0	None — indoors, any weather	Background/SNR constants; gphoto2 capability truth; saturation values; focus-reach answer. **Unblocks D1.**
T1	One clear night, own gear + Barlow	TED (RB-6); FQE without positions (RB-2); COC in parallel (RB-5).
T1b	Any night, even cloudy	COC alone (RB-5). **The cheapest data in the plan.**
T2	An electronic focuser — bought, or included with the loaner	FOP, FDF, FQE with full provenance.
T3	Rental window	Exposure bandit; tracked negatives for TED; long-exposure FQE; plate-solved frames for TME.
T4	Filter wheel	FOR. **Descoped.**

8.2 The critical path

1. **BP-0** — compile the daemon on the Mac. Nothing downstream is real until this passes. Roughly 1,500 lines of Path B have never been through a compiler.

2. **BP-2** — photon transfer curves for all five bodies. Indoors, one evening each, weather-independent.
3. **Fix D1** — dataset builder reads `saturation_adu`; write the measured values into config. Half an hour of code, and it gates every focus label.
4. **BP-1, BP-3** — gphoto2 truth table and focus-reach test. These determine which bodies can be planned around at all.
5. **Build the sidecar importer** (G2/G4). Without it, R4 — the cheapest and most parallel rig — produces orphaned files.
6. **Decide the focuser question** before booking the rental. If the loaner includes an INDI/Alpaca focuser, T2 and T3 merge into one window and you save both the money and a month. If not, buy one — FOP and FDF are the models the entire §7 control narrative rests on, and neither is collectable without it.
7. **Start RB-5 immediately** and never stop. It costs nothing, it runs on nights that are useless for everything else, and COC needs more distinct nights than any other model.

Sequencing note on the rental. The rental window is the scarce resource, so it should be spent last, not first — after BP-0/BP-2 have proven the software path, after the importer exists, and after RB-2/RB-6 have shaken out the field procedures on gear you can re-shoot for free. Arriving at a 14-day loan with untested capture code is how the window gets spent on debugging. Aim to book it around a new moon, and pad for shipping.

8.3 Risk register

Risk	Severity	Mitigation
Path B does not compile or crashes on first RAW	High	BP-0 early; keep Path A (streaming) as the fallback for all FQE work — it is the mature path.
libuvc cannot open the SV105 on macOS	High for R1/R2	BP-1. Fallback: Linux VM with USB passthrough, or AVFoundation capture + sidecar import.
DSLR will not reach prime focus on the Z114	Medium	BP-3, daytime. Fallback: SV105 only for R1/R2; DSLRs go to R4 and R5.
R50 / G95 cannot be tethered	Medium	BP-1. Fallback: assign them to open-loop campaigns (R4) where the shutter is on an intervalometer.
Loaner has no electronic focuser	High for FOP/FDF	Ask before booking (§3.3, question 3). Fallback: dial indicator (RB-2) or purchase.
Weather — Piedmont NC summer	High	T1b (COC) converts bad nights into data. T0 is weather-independent. Front-load both.
Undersampling at native focal length	Medium	2× Barlow, treated as mandatory for R2/R3.
Focus runs contaminated by unlogged disturbances	Medium	Disturbance block on the field card; drop the run rather than clean it.

Part IV Where the data comes from

9 Sourcing routes: the decision frame

9.1 The rule that governs this whole part

The own-gear track is the critical path. Everything else is optional upside, and nothing else may block it.
Every route in this part except the first has a response latency measured in weeks and an outcome outside your control — an email that may not be answered, a loan programme that may not have a focuser, a rental window that may not fall on a clear moon. Plan the build and the observing campaign as though none of them will come through. If one does, it is a bonus track that runs in parallel.
The failure mode this rule exists to prevent is spending September waiting for a reply instead of printing a bracket.

9.2 What each route can and cannot supply

Rows are the nine models of Part III. A tick means the route can produce training-grade data for that model; a tilde means partial or unverified; a cross means it structurally cannot.

Model	Own gear R1–R4 + DIY	UNCG/TCO free	LCO free	Skynet $65	iTelescope $50/hr	Seestar $349+	GoTo mount $200
FQE (7.6.1)	✓✓	~	×	×	×	~	×
FOP (7.6.2)	✓✓	✓✓	×	×	×	✓	×
FDF (7.6.3)	✓✓	✓✓	~	~	~	✓	×
COC (7.6.4)	✓✓	×	×	×	××	×	×
TED (7.6.5)	✓✓	~	~	~	✓✓	✓	×
Background (7.6.6)	✓✓	✓	✓	✓	✓	✓	×
Bandit (7.6.7)	×	~	~	~	✓✓	✓	✓
FOR (7.6.8)	×	~	✓	✓	✓✓	×	×
TME (7.6.9)	~	✓✓	✓	✓	✓	~	×

Three things fall out of that table immediately.

1. **Own gear is the only route that can produce COC data at all.** Every remote and institutional option closes the dome when it is cloudy. The Cloud/Obstruction Classifier is structurally uncollectable anywhere except a camera you personally point at a bad sky.
2. **Own gear plus a DIY focuser covers six of nine models.** It is not a fallback. It is the main plan.
3. **The paid routes buy exactly the two models that were already descoped** — the exposure bandit and the filter recommender — because both need hardware you do not have and neither is on the critical path.

9.3 Switching triggers

Rather than choosing a route now, define the condition under which each becomes worth activating.

Route	Activate when
UNCG / TCO	Immediately — it costs one email and runs in parallel.
LCO archive	Immediately — public data, no login, no application.
LCO Global Sky Clubs	If the Durham club will sponsor an application.
Skynet	If $65 is affordable and you want a scriptable API to write a driver against.
Seestar S30/S50	If the DIY focuser build fails or slips past October, *or* if you specifically want an end-to-end Alpaca integration rig.
iTelescope	Only after FQE/FOP/FDF are done and you want FOR and the bandit. Not before.
GoTo mount	Never, on this budget. See §11.6.
Raleigh loaner	Never. $25 plus residency proof plus in-person pickup for gear worse than TCO's.

10 The UNCG track

10.1 Three College Observatory

TCO is operated jointly by UNCG, NC A&T, and Guilford College. It holds a **0.81 m (32") Ritchey-Chrétien on an equatorial mount** — the largest telescope in North Carolina — at a dark-sky site at 5106 Thompson Mill Road, Graham, roughly 33 miles and 45 minutes from Greensboro, and about midway between Greensboro and Chapel Hill.

The instrument that matters is a **fibre-fed eShel échelle spectrograph**, $R \approx 12{,}000$, covering 3800–7900 Å continuously.

Public nights run roughly monthly from late September to June; the observatory's own page notes that summer months are too cloudy to be worth scheduling. That sets the season boundary for anything involving the site.

Prof. Anatoly Miroshnichenko is the Director of TCO and of the UNCG Planetarium, and has already replied to an initial enquiry saying there is data to work on. That single fact outranks every paid option in this part.

10.2 What his group actually works on

Verified from published sources, not inferred:

- Hot stars with circumstellar material, the B[e] phenomenon, and binary systems. Co-author on *Galaxies* **13**, 67 (2025), affiliated to UNCG Physics & Astronomy.
- A **long-term binary spectroscopy programme at the TCO 0.81 m running since 2013**. One recent paper alone uses 71 TCO échelle spectra spanning 2014–2025; TCO spectroscopy of Polaris Aa covers 2015–2024.
- Photometry used for **candidate selection**: the 2025 paper builds criteria from optical and near-infrared colour indices, finds ∼40 candidate B[e] objects, then uses spectroscopy to classify them.

10.3 What to ask for, in order

Ask 1 — the archive project (his benefit)

The select-then-confirm loop in the 2025 paper is precisely the setting where a cross-modal embedding helps: hand-built colour cuts are replaced by ranking against confirmed examples in a learned space over photometry, spectra and metadata. AstroM3's published weights already exist, so this is an application, not a training project.

The reason this is a good proposal rather than a speculative one is that it is **falsifiable against work already done**: their existing candidate list and spectroscopic follow-ups form a ready-made validation set. If the method does not recover them, it does not work.

Ask 2 — the autofocus and engineering logs (your benefit)

Any observatory running ACP, MaxIm, NINA or TheSkyX writes (focuser position, HFD/FWHM, temperature) triples every night, plus bias, dark and flat libraries. These are treated as throwaway operational exhaust. They are **exactly the FOP and FDF training set**, and a decade of them may already exist on disk.

This is the highest-value-per-effort request in the entire document: it costs the observatory one zip file and could supply, in a single transfer, the hundreds of focus runs that §7.2 says will otherwise take twenty nights to accumulate.

Be honest that this serves your project, not theirs. Disguising it as collaboration is both wrong and transparent.

Ask 3 — engineering time (small, later)

Not "may I observe" but "may I run a defocus sequence on a bright star during engineering time." Focus sweeps need fifteen minutes and a bright star — not dark time, not photometric conditions. Offer explicitly to take the nights nobody else wants.

10.4 The second contact

Dr. Alicia Aarnio (UNCG Physics & Astronomy) works on stellar magnetic fields and young stars using large datasets and computational methods, and runs an undergraduate research group. She is the more natural contact for the software and ML side specifically, and has an existing student pathway.

10.5 What the UNCG track does and does not unblock

Unblocks	FOP and FDF, *if* the autofocus logs exist and are shared. TME, via the échelle archive. Background/SNR, via their calibration frames.
Does not unblock	COC — they do not observe through cloud. FQE — unless engineering time is granted, and even then the runs are theirs, not repeatable on demand.
Does not replace	The DIY focuser. Data you cannot re-collect at will is not a substitute for an instrument you control.

11 Remote networks and purchased hardware

11.1 iTelescope

Commercial remote network. Control is by uploading an ACP plan text file; the complete directive set is `#FILTER`, `#COUNT`, `#INTERVAL`, `#BINNING`, `#SUBFRAME`, `#DITHER`, `#AUTOFOCUS`, `#AFINTERVAL`, `#FILTEROFFSETS`, `#DEFOCUS`, `#PLATESOLVE`, `#TRACKON`/`#TRACKOFF`, `#DARK`, `#BIAS`, `#EXPRESS`.

There is no directive that commands an absolute focuser position. `#DEFOCUS` is documented as "defocus the image slightly" with no parameter — one binary step. No sweep is possible, so FQE and FOP get nothing. There is also no ASCOM/INDI/Alpaca surface exposed to the user, so the daemon cannot drive it at all.

What it *is* good for: `#TRACKON` performs orbital tracking, which their documentation notes produces "trailed stars and a stationary minor planet" — deliberate, rate-controlled trailing on a professional rig, which is excellent TED data. Real L/R/G/B/Hα/OIII/SII wheels make FOR collectable. Long tracked exposures with varied moon and airmass make the bandit collectable. And moon discounts make full-moon time cheap, which is exactly when sky-brightness data is wanted.

Cost: 1 point = US$1. Their worked example is 104 points for two hours of session time on T21 at the Plan-40 tier, i.e. ~**$52/hour**; rates fall on larger plans. Billing is per-session or per-exposure, with a typical "system efficiency" near 0.7 — about 42 minutes of exposure per hour of session.

Education accounts exist, but the application requires an institutional email address and explicitly rejects public domains such as Gmail. A UNCG address would be the lever.

11.2 Skynet (UNC Chapel Hill)

Roughly twenty optical telescopes across four continents plus the Green Bank radio telescope, operated from UNC-Chapel Hill. Access is normally restricted to contributing institutions, but is open to anyone taking the tuition-free, self-paced *Astronomy with Skynet* courses, for which WebAssign charges about $65.

The reason it matters more than iTelescope: **Skynet has a public, token-authenticated API** for submitting observations. That is something a daemon driver could be written against, which makes it Achernar-shaped in a way a web form is not.

11.3 Las Cumbres Observatory

- **Global Sky Partners** awards up to 1500 hours per year to educational programmes; **Global Sky Clubs** gives 10 hours on the 0.4 m network to school astronomy clubs.
- The **science archive is public** — data without a proprietary period is downloadable with no login, and proprietary data becomes public after 12 months. Access is by web UI or a JSON API.
- Pipeline-processed data is available as little as 15 minutes after the shutter closes.

The LCO public archive is the only route in this part that requires no application, no payment, no email, and no waiting. It is real FITS with real headers, available today, and it is enough to exercise the ingestion path, the background estimator, and the TME bookkeeping while the sky is cloudy.

11.4 Smart telescopes

The ZWO Seestar S50 ($499, discontinued early 2026) and S30 ($349) expose **camera, mount and focuser over native Alpaca** in recent firmware; the open-source `seestar_alp` project provides an alternative controller. An S50 Pro is expected mid-to-late 2026 at $599–799.

Because the daemon's `alpaca` driver is always compiled and needs no SDK, a Seestar is a genuine **drop-in end-to-end integration rig** — tracking mount, camera, and absolute-position focuser, all reachable with no new code.

Its limits are optical, not electrical: 50 mm at f/5 (250 mm focal length) gives roughly 2.4 "/px, badly undersampled, and the optics are fixed. Treat it as an *integration* instrument and a fallback if the DIY build slips — not as an FQE rig.

11.5 Focusers

Option	Cost	Note
Dial indicator	~$15	Manual absolute position to 10 µm. No automation.
ESP32 + myFP2ESP32	~$40	**Recommended.** Speaks Alpaca, ASCOM, INDI and TCP/IP natively.
ZWO EAF	$179	`indi_asi_focuser`. No build risk.

Why the DIY build must be ESP32 and not Arduino. `OpenLiveStacker/src/focuser.cpp` is 46 lines and its driver list returns exactly one entry, `{{"alpaca:remote","auto"}}`. OLS implements no Moonlite driver, no USB-serial driver and no INDI focuser driver — despite `hardware/focuser.rs` carrying `FocuserDriver` variants that map to `"moonlite"` and `"usb"`. Those names point at drivers that do not exist, which is what `capabilities.rs` reports as `disabled`.
An Arduino speaking the Moonlite protocol therefore requires writing a new OLS C++ driver and threading it through the bridge ABI. An ESP32 speaking Alpaca connects to `alpaca_focuser.cpp`, which is already written and already compiled, with **zero new C++ or Rust**.
Secondary benefit: the Alpaca focuser interface carries `Temperature` and `TempComp` as standard properties, so a DS18B20 on the same board closes **D5** at the same time as **D3**.

11.6 Mounts — and why not to buy one

A $200 GoTo mount solves tracking. **Tracking is not the blocker.** Consult the table in §9.2: a mount unlocks the exposure bandit and nothing else, and the bandit was already descoped. A $40 focuser unlocks FOP and FDF, which are the two models the entire §7 control narrative rests on.
Sub-$300 GoTo mounts are additionally alt-azimuth, which introduces field rotation, and none ship with a drivable focuser. If tracked time is later needed, TCO has an equatorial 0.81 m and LCO gives hours away.
Spend on the focuser. Do not buy a mount.

11.7 Standing recommendation

1. Build the DIY focuser and observe on own gear. This is the plan.
2. Email UNCG in parallel, expecting nothing on any particular timescale.
3. Pull LCO archive data this week — it is free and needs no permission.
4. Consider Skynet's $65 for API access worth writing a driver against.
5. Hold iTelescope in reserve for FOR and the bandit, after the focus stack is finished.
6. Buy no mount.

Part V Field and bench procedures

Every procedure below states, in order: **purpose**, **preconditions**, **equipment**, **numbered steps**, **what is recorded**, **abort criteria**, and **acceptance**. A procedure that is aborted still produces a log entry — an aborted run is evidence, and the `capture_attempts` table exists so that failures are counted rather than vanishing.
Nothing here requires the daemon to be working. Where a step names a daemon endpoint, a manual equivalent is given.

Appendix A — Bench procedures (indoors, weather-independent)

BP-0 — First compile and ABI verification

Purpose. Determine whether the still-capture and RAW-decode path (roughly 1,500 lines written but never compiled) actually builds and runs.

Preconditions. Mac with Xcode command line tools, Homebrew, Rust toolchain.

1. Install the optional native libraries. Each one flips a capability bit that `hardware/capabilities.rs` reports at runtime:

```
brew install libraw libgphoto2 libusb opencv libtiff
# optional, enables the INDI camera/mount/focuser drivers:
brew install indi
```

2. Build, capturing the log — the CMake summary lines matter as much as the exit code:

```
cd achernar-daemon
cargo build 2>&1 | tee /tmp/build.log
grep -Ei "libraw|gphoto|uvc|indi|Decode:" /tmp/build.log
```

3. Expect signature mismatches from the capture work. **Fix them against a clean tree first** and do not tangle them with new features.
4. Run and read the startup banner. It should print the OLS Bridge ABI version and whether LibRaw was found.

```
cargo run 2>&1 | head -40
```

5. Query the capability report and save it. This file is the ground truth for every planning decision in Part I, and it should be re-saved whenever the build environment changes:

```
curl -s localhost:8080/drivers | tee docs/capability-report-$(date +%F).json | python3 -m json.
    tool
```

6. Verify D2: confirm whether `camera_config.saturation_adu` reaches the still-capture path, not only `start_stream`. Search `src/api/camera.rs` for the capture handler and check what it passes into the quality code. If it does not, wire it through now.
7. Run the test suite: `cargo test`. Then run the Python side: `python -m pytest ml/tests/`, which exercises the dataset builder and the FQE pipeline against synthetic fixtures.

Recorded. `docs/capability-report-YYYY-MM-DD.json`; the build log; a note of every signature fix made.

Abort criteria. None — this must be completed.

Acceptance. `cargo build` and `cargo test` pass; the capability report shows `gphoto2: compiled = true`; the startup banner confirms LibRaw.

BP-1 — Camera capability probe (per body)

Purpose. Establish, per body, whether gphoto2 can *trigger* an exposure — not merely list files — and which exposure and ISO values are actually settable. This determines which bodies can be planned around.

Preconditions. BP-0 passed. **Turn off the camera's Wi-Fi** and any companion-app pairing; on Canon bodies set USB connection mode to PC/PTP, not MTP or "smartphone".

Time. 20–30 minutes per body. Repeat for all five.

1. Connect over USB. On macOS, quit Photos, Image Capture and any auto-launching importer first — they claim the device and gphoto2 will report it as busy.
2. Detect and dump abilities:

```
gphoto2 --auto-detect
gphoto2 --abilities
gphoto2 --summary
```

   In `-abilities`, the decisive line is the capture operations list. If it does not include image capture, the body is download-only.
3. Enumerate the settable configuration tree and save it — this is what `hardware/dslr_controls.rs` parses with `parse_shutter_seconds` and `parse_iso`:

```
gphoto2 --list-config > docs/gphoto-config-<body>.txt
gphoto2 --get-config /main/capturesettings/shutterspeed
gphoto2 --get-config /main/imgsettings/iso
gphoto2 --get-config /main/imgsettings/imageformat
```

   Record whether `bulb` appears among the shutter choices and whether `/main/actions/eosremoterelease` exists (Canon) — bulb is required for exposures beyond 30 s.
4. Set RAW-only output, a known shutter and a known ISO, then read them back. *Reading back is the point* — the daemon's capture path applies and re-reads for exactly this reason, and a body that silently ignores a set is worse than one that refuses it:

```
gphoto2 --set-config imageformat=RAW
gphoto2 --set-config shutterspeed=1/60 --set-config iso=800
gphoto2 --get-config shutterspeed --get-config iso
```

5. **The decisive test.** Attempt three remote captures with download, and confirm three *new* files with three *distinct* timestamps:

```
for i in 1 2 3; do
  gphoto2 --capture-image-and-download --filename "probe_${i}.%C"
done
ls -l probe_*
```

6. Time the round trip (trigger to file on disk). This sets the realistic cadence for every sweep in Appendix B — a 6-second round trip turns a 21-position, 5-frame sweep into 10 minutes of pure transfer.
7. Repeat steps 4–6 with `shutterspeed=bulb` if available, using a 5-second bulb exposure, and confirm the resulting file's actual exposure in EXIF.
8. Verify LibRaw can decode the result:

```
# any LibRaw sample tool, or:
python3 -c "import rawpy,sys; r=rawpy.imread(sys.argv[1]); print(r.raw_image.shape, r.raw_image.
    dtype, r.black_level_per_channel, r.white_level, r.raw_pattern)" probe_1.CR3
```

   Confirm the reported white level. **This number is a candidate for `saturation_adu`, but do not trust it** — BP-2 measures it. The metadata value is a manufacturer hint and is frequently wrong by a few hundred ADU.

9. For the SV105, the equivalent probe is a UVC one:

```
# does the OS expose it, and can libuvc claim it?
system_profiler SPCameraDataType
# then attempt capture through the daemon's uvc driver and check
# the reported stream format is 8-bit
curl -s localhost:8080/camera/formats | python3 -m json.tool
```

   Confirm the reported container is 8-bit, so that `resolve_saturation_adu` returns `Some(255)` and the saturation gate can fire.

Recorded. One row per body in the capability table below, plus the saved config dumps.

Body	detect	set shutter	readback	capture	bulb	round trip	LibRaw
T3i							
T6S							
M50							
R50							
G95							
SV105	UVC path: opens / 8-bit confirmed / max exposure confirmed						

Abort criteria. If `-capture-image-and-download` fails 3 of 3, stop probing that body and mark it **open-loop only**. Do not spend more than 30 minutes per body — the fallback (intervalometer + sidecar import) is perfectly adequate for COC, background and trailing work.

Acceptance. At least two bodies pass full remote capture. If none do, the entire closed-loop plan collapses to the SV105 and this must be discovered now, not on a rental night.

BP-2 — Photon transfer curve, read noise, and saturation

Purpose. Measure, per (body, ISO), the conversion gain g in e^-/ADU, the read noise σ_{read} in e^-, the black level, and the linear saturation point in ADU. These are the constants the Background/SNR estimator needs (§7.6) and the value D1 needs.

Preconditions. BP-1 passed for this body. **Indoors. Any weather.**

Equipment. Body; lens or body cap; a uniform light source — a white laptop or tablet screen at fixed brightness, viewed slightly defocused, is more than good enough; a dark room; tripod or stable surface.

Time. 45–60 minutes per body, per ISO. Do at least ISO 100, 400, 1600, 3200.

1. **Bias frames.** Body cap on, lens off, in a dark room. Set the shortest available shutter (1/4000 s). Capture **20 frames**. Do not let the sensor heat — pause 5 seconds between frames.
2. **Dark frames.** Same, but at each exposure you intend to use in the field (1 s, 10 s, 30 s, and bulb 120 s if applicable), **10 frames each**. These also give you the dark-current rate, which is strongly temperature dependent — record the EXIF sensor temperature for every frame.
3. **Flat pairs.** Point at the uniform source. At each of **12 illumination levels** spanning roughly 5 % to 105 % of full scale (vary shutter speed, not screen brightness — brightness controls on screens are often non-linear and PWM-modulated), capture **2 frames**. Twelve levels × 2 = 24 frames.
   Use exposures ≥1/100 s to avoid screen PWM banding, and check one frame for banding before committing to the whole ladder.
4. **Compute.** For each level, using the central 500×500 region of one Bayer channel only (do *not* mix channels — they have different sensitivities):
   - mean signal $\bar{S}$ and the variance of the *difference* of the pair, $\sigma^2_{(S_1-S_2)}$;
   - the same for the bias pair;

- $g = [(\bar{S}_1 + \bar{S}_2) - (\bar{B}_1 + \bar{B}_2)] / [\sigma^2_{(S_1 - S_2)} - \sigma^2_{(B_1 - B_2)}]$.

Differencing is what removes fixed-pattern non-uniformity, which would otherwise dominate the variance and make g come out far too small.

5. **Plot** σ^2 against $\bar{S}$. The linear region has slope $1/g$. **The saturation point is where it turns over**, not the maximum value in the file.
6. **Read noise.** $\sigma_{\text{read}} = g \times \sigma_{\text{bias}}$, where σ_{bias} is the standard deviation of the bias difference frame divided by $\sqrt{2}$.
7. **Black level.** The mean of the bias frames. Canon bodies typically carry an offset near 2048 ADU for 14-bit output. **Measure it, do not assume it** — it varies by body and sometimes by ISO.
8. **Write the results** to `docs/device-profiles/<body>.json` and set `[camera] saturation_adu` in the daemon config for the body in use. Follow the config file's own warning: a wrong value is worse than none.

Recorded. Per (body, ISO): g, σ_{read}, black level, saturation ADU, full-well in e^-, dark current in e^-/px/s at the measured sensor temperature.

Abort criteria. If the PTC is not linear over at least a decade of signal, something is applying a non-linear transform — almost always a highlight-tone curve or the wrong RAW decode path. Stop and fix; the numbers are meaningless otherwise.

Acceptance. g within a factor of 2 of the manufacturer's published value where one exists; σ^2-vs-$\bar{S}$ linear ($R^2 > 0.99$) below saturation.

BP-3 — Prime-focus reach and backfocus (daytime)

Purpose. Determine whether each camera can physically reach focus on the Z114, with and without the Barlow. Many tabletop Newtonians place the focal plane inside the drawtube, which no DSLR can reach.

Time. 15 minutes. **Daylight. Do not point anywhere near the sun.**

1. Set up indoors, aimed through a window at a high-contrast object at least 1 km away — a distant roofline or antenna.
2. Insert the SV105 at prime focus. Rack the focuser fully in, then out slowly, watching the live stream. Record whether focus is achieved and, if so, the drawtube extension in mm at focus.
3. Repeat with each DSLR on a T-ring + 1.25" nosepiece.
4. Repeat both with the 2× Barlow inserted — a Barlow moves the focal plane *outward*, which usually rescues a body that could not otherwise reach focus. This is a second, independent reason to own one.
5. If a DSLR cannot reach focus even with the Barlow, record the shortfall in mm. Remedies, in order of preference: use the Barlow permanently; fit a low-profile focuser; move the primary mirror up the tube (last resort — it changes the optical alignment).

Recorded. A 6-row table: camera × {native, Barlow} → {reaches focus?, drawtube mm at focus, shortfall mm}.

Acceptance. At least one camera reaches focus in at least one configuration. If none do, R1–R3 are void and every telescope-based campaign depends on the rental.

BP-4 — Plate scale and orientation by solving

Purpose. Measure the true arcsec/px and field rotation. **Do not trust the nominal focal length** — a Barlow's true amplification depends on its spacing from the sensor and is routinely 10–20 % off its label, and the trail lengths in RB-6 are computed in arcseconds but labelled in pixels.

Preconditions. One clear night, 20 minutes. Astrometry.net installed locally with the appropriate index files (4200-series for wide fields, 4100-series for narrow).

1. Point at a moderately rich star field well away from the Milky Way plane — too many stars is as bad as too few.
2. Capture 3 frames at the shortest exposure that shows ≥30 stars.
3. Solve:

```
solve-field --scale-units arcsecperpix --scale-low 0.2 --scale-high 3.0 \
            --downsample 2 --no-plots --overwrite frame.fits
```

4. Record from the WCS: pixel scale, field rotation, and the solved centre.
5. Repeat for every optical configuration you intend to use (native, Barlow, each camera). **Each combination has its own scale.**

Recorded. `docs/plate-scales.json`, keyed by rig ID.

Acceptance. Solved scale within 15 % of the computed value in §3.4. A larger discrepancy means the assumed focal length or pixel pitch is wrong — find out which before collecting anything that depends on it.

BP-5 — Focuser step calibration (only if motorised)

Purpose. Populate `focuser_profiles.steps_per_micron` (migration 005) and the critical focus zone, both of which the FDF trigger ($0.5 \times$ CFZ) and the FOP normalisation depend on.

1. Command a known large move (e.g. 1000 steps) and measure the drawtube displacement with a dial indicator. steps/μm $= 1000/\Delta x_{\mu m}$.
2. Measure backlash: approach a position from each direction and record the difference. Write it to `focuser_profiles`; `focus/backlash.rs` consumes it.
3. Compute the CFZ from the focal ratio: CFZ $\approx 4.88\,\lambda N^2$ in µm, with $\lambda = 0.55\,\mu$m and N the focal ratio. For f/3.95 this is $\approx 42\,\mu$m; for f/10, $\approx 268\,\mu$m. Convert to steps with the measured scale.
4. **Sanity check.** The step size should be smaller than CFZ/5, or the focuser cannot resolve best focus regardless of how good the model is.

Recorded. steps/µm, backlash in steps, CFZ in µm and in steps, per rig.

Appendix B — Observing runbooks

RB-1 — Calibration frame library

Purpose. Bias, dark and flat libraries per (body, ISO, exposure, temperature bin). Required for background estimation and any later calibration; also the substrate for BP-2.

When. Refresh whenever the body changes, the ISO set changes, or ambient temperature moves by more than 5 °C from the library's bin.

Type	Count	Conditions	`capture_type`
Bias	50	shortest shutter, cap on, dark room	`bias`
Dark	20 per exposure	matched exposure, ISO, temperature bin	`dark`
Flat	20 per filter/config	even illumination, ∼50 % full scale	`flat`
Flat-dark	20	matched to the flat exposure	`dark`

1. Shoot bias and dark at the end of the session, while the equipment is still at ambient. Darks shot warm in the afternoon do not match lights shot cold at 2 a.m.

2. Shoot flats at dusk against a twilight sky, or with a white t-shirt over the aperture and an even light source. Target a mean around half of the *measured* saturation from BP-2 — not half the container ceiling.
3. Record the sensor temperature (EXIF) for every calibration frame.
4. Set `frames.capture_type` correctly on ingest (gap G5). If the daemon does not yet set it, the sidecar importer must.

Abort criteria. Any light leak visible in a bias frame — reshoot in a darker room, and check the viewfinder eyepiece is covered on DSLRs.

Acceptance. Bias frames have a flat histogram centred on the black level from BP-2 with no gradient; flats vary by $< 20\,\%$ corner to centre beyond expected vignetting.

RB-2 — Focus sweep with a manual focuser (R2 / R3)

Purpose. Produce FQE training runs — and, with the dial-indicator modification, low-precision FOP runs — without a motorised focuser.

Preconditions. BP-2, BP-3, BP-4 complete. Clear sky, seeing not terrible, wind below about 10 km/h.

Equipment. Z114 + 2× Barlow + SV105 (or DSLR). A dial indicator (0.01 mm) clamped so its plunger rides the drawtube, or failing that a printed paper mm scale taped along the drawtube. Red headlamp. Field card.

The dial-indicator modification. A $15 dial indicator clamped to the focuser body with its plunger against the drawtube gives absolute position to 10 µm. At f/3.95 the CFZ is $\approx 42\,\mu$m, so 10 µm resolves the focus zone into four steps — coarse, but *real absolute position*, which is the difference between an FQE-only run and an FOP-eligible one. Record the indicator reading in `frames.focuser_position` in units of 10 µm and document the unit in `acquisition_json`.

1. **Setup.** Level the table. Let the telescope reach ambient for ≥30 minutes — a warm mirror produces tube currents that ruin every HFR measurement. Record the start time and ambient temperature.
2. **Target.** A field with 20–60 stars of moderate brightness. Avoid a single very bright star: it saturates while the field stars are still undetected, and the saturation gate will reject exactly the star you were aiming at. **A loose open cluster is ideal.**
3. **Exposure.** Set so the brightest stars peak at 50–70 % of the measured saturation *when in focus* — because they will be far dimmer when defocused, and you must not clip at the centre of the sweep. For the SV105 this will be in the 50–300 ms range.
4. **Find visual best focus** and record the indicator reading as x_0.
5. **Define the sweep.** 21 positions, symmetric about x_0, step size $\approx$ CFZ/3 $\approx 14\,\mu$m — so $x_0 \pm 140\,\mu$m end to end. If the V-curve does not clearly turn over at both ends, widen the range and redo.
6. **At each position, in order:**
   a. Move the focuser **always in the same direction** — start fully at one end and march to the other. Never back up mid-sweep; backlash will corrupt the position axis.
   b. Wait 3 seconds for vibration to settle.
   c. Record the indicator reading.
   d. Capture **5 frames**.
   e. Note star count from the live display if available.

7. **Log the run** on the field card: run ID, start/end time, x_0, step size, direction, ambient temperature at start and end, target, altitude, and anything you touched.
8. **Repeat** 3 times per night minimum. Vary the target field between runs — three sweeps on the same field are more correlated than three on different fields, and correlated groups do not help a grouped split.
9. **One run in three should be a reversal run** (see §7.2): start 70 µm inside the V on the wrong side, move away from focus for four positions, then reverse and complete the sweep. This is the only source of variation for the FOP direction and history features.

Recorded. Per run: 21 positions × 5 frames ≈ 105 frames. Per position: indicator reading, timestamp, exposure, gain/ISO, star count. Per run: ambient temperature at start and end, target, altitude, seeing impression, disturbances.

Abort criteria.

- Cloud crossing the field — abort, log the run as `aborted`, restart.
- Star count < 8 at the centre of the sweep — the field is too sparse.
- Ambient temperature drifts $> 0.5\,°C$ within the run — keep the data but flag it; it belongs to RB-4, not here.
- Wind gusts visibly shaking the image — abort.

Acceptance. The HFR-vs-position curve is a clean V or U, monotonic on each side, minimum interior to the sweep, with ≥15 stars at the minimum and ≥8 at both ends.

Yield. ∼15 min per run, ∼10 runs per productive night. **v1 (60 runs) is 6 clear nights; v2 (150 runs) is 15.**

RB-3 — Focus sweep with a motorised focuser (R5)

Purpose. Full-provenance FQE runs plus FOP rows and FDF pairs. This is the procedure that produces all three focus models at once and is therefore the best use of a rental night.

Preconditions. BP-5 complete; focuser reachable through INDI or Alpaca; `POST /focus/run` works.

1. Setup and thermal equalisation as RB-2 steps 1–3.
2. Verify the daemon can read and command position:

```
curl -s localhost:8080/focuser/position
curl -s -X POST localhost:8080/focuser/moveto -d '{"position": 12000}'
```

3. **25 positions**, step size CFZ/4 in steps from BP-5, **3 frames** per position, always approaching from the same direction with backlash compensation enabled.
4. After each run, record the fitted best position and the ambient temperature — this pair is one FDF observation.
5. **Vary deliberately across runs:**
   - **Field position:** centre, and two fields near the edge of the usable field. Off-axis aberration changes the PSF shape and the model must not learn that the centre crop is the only valid one.
   - **Airmass:** one field near zenith, one at ∼2 airmasses.
   - **Starting offset:** alternate between starting near focus and starting at ±150 steps, to populate the balanced-sampler buckets of §7.6.2.
   - **Direction:** alternate inward-first and outward-first.
6. Log every run to `focus_runs` with a correct `rig_id`.

Yield. ∼8 min per run with a motor. **∼25 runs per night**, which is where the FOP budget becomes achievable — 200 runs is 8 nights rather than 20.

RB-4 — Thermal drift night (FDF)

Purpose. Fit the per-tube temperature coefficient α in steps/°C, and the drift rate $\dot{p}$.

Preconditions. A focuser with position readout. A thermometer. **Choose a night with a forecast temperature drop of ≥8 °C** — clear, dry, post-frontal nights are ideal, which conveniently are also good seeing nights.

1. Start **at dusk**, while the air is still warm, and run until the pre-dawn minimum. The span is the whole point; a 3-hour session in the middle of the night has almost none.
2. Do **not** let the telescope reach thermal equilibrium first — for once, you want the cooling transient.
3. Run a full autofocus **every 30 minutes**, on the same field if possible, and **do not touch the focuser between runs**.
4. Record at every run: ambient temperature, sensor temperature, focuser best position, time.
5. Additionally log ambient temperature every 15 minutes regardless, so the temperature series is denser than the focus series.
6. Continue for ≥6 hours, giving ≥12 runs.

Recorded. A table of (time, T_{ambient}, T_{sensor}, p_{best}, run_id). This is the entire FDF dataset for that tube.

Abort criteria. Cloud for more than 30 minutes — resume and note the gap; do not interpolate across it. If the focuser is bumped, the night is over for FDF purposes (though the individual runs remain valid FQE data).

Acceptance. ≥10 successful runs spanning ≥6 °C, and p_{best} versus T approximately linear. A scatter plot with no visible trend means either the temperature range was too small or the focuser has more backlash than drift — check BP-5.

Repeat. Three nights per tube. α is a property of the tube, not the camera; a loaner's α does not transfer to the Z114.

RB-5 — Cloud and obstruction campaign (R4)

Purpose. COC training data. **The cheapest and most parallelisable campaign in the plan.**

Preconditions. None. No telescope, no daemon, no rental, no clear sky.

Equipment. Any body + widest lens; tripod; intervalometer or the body's built-in interval timer; spare battery (this runs all night); a phone for cross-check photos; the field card.

1. Set up pointing at a fixed altitude/azimuth — roughly 45° altitude, and choose the azimuth deliberately. Run some nights aimed at the Durham/Chapel Hill light dome and some aimed away, so the model sees both. Record which.
2. Manual mode, fixed exposure, fixed ISO, fixed focus at infinity, fixed white balance. **Nothing may auto-adjust** — an auto-exposure change looks exactly like a transparency change and will teach the model the wrong thing. RAW + JPEG.
3. Interval: **one frame every 30 seconds**, all night. At 8 hours that is 960 frames per night.
4. **Label every 10 minutes** on the 5-point scale below. Record transitions to the minute.
5. Take a phone photo of the sky at each label as an independent adjudicator.
6. Note obstructions explicitly: tree, roofline, power line, aircraft, contrail, car headlights, dew on the lens.
7. Leave it running when you go inside. This campaign does not need supervision, only labels — set a 10-minute timer and step out.

Code	Class	Criterion	Binary
0	clear	no cloud visible; Milky Way structure visible if moonless	0
1	thin	cirrus or haze; stars visible through it; halo around bright stars	0
2	moderate	distinct cloud over ∼25–60 % of the frame; some stars hidden	1
3	heavy	>60 % covered; only the brightest stars	1
4	overcast	no stars visible	1

Recorded. Frames + a label timeline + moon altitude/illumination (computable) + azimuth + humidity + phone photos.

Abort criteria. Rain — bring the camera in. Dew on the lens — photograph it once deliberately as a hard negative, then dry it and note the timestamp of the wipe.

Acceptance. ≥500 labelled frames per night. Over the campaign, ≥800 frames in each of the three rarer classes.

Class balance strategy. After each night, tally per class and pick the next night to fill the thinnest class. In this climate, class 1 (thin cirrus) is usually the hardest to get deliberately and class 4 (overcast) the easiest and least useful — cap overcast at about 15 % of the set.

RB-6 — Trailing ladder (R1 / R2)

Purpose. TED training data with **exactly computed** elongation labels, exploiting the Z114's lack of tracking.

Preconditions. BP-4 — the plate scale must be solved, not assumed, because the label is computed in arcseconds and consumed in pixels.

The label. For an untracked exposure of t seconds at declination δ, the trail length is

$$\ell_{\text{arcsec}} = 15.041\,\cos\delta\,t, \qquad \ell_{\text{px}} = \ell_{\text{arcsec}}/s,$$

with s the solved plate scale. The expected ellipticity of a star of seeing-limited FWHM w (in px) trailed by ℓ_{px} is approximately $e = 1 - w/\sqrt{w^2 + \ell_{\text{px}}^2}$, which is what the classical estimator will report and what the CNN must learn to agree with in the ambiguous band.

1. Choose **3 fields** at declinations near 0°, +40° and +70°. $\cos\delta$ is 1.00, 0.77 and 0.34 — so the same exposure gives three different trail lengths and the label grid is 2-D.
2. At each field, run the exposure ladder below. **Sample densely between 0.05 and 0.25 s** — that is where the CNN's ambiguous band $0.10 < e < 0.20$ falls, and a linear ladder under-samples it badly.

t (s) ∈ {0.017, 0.033, 0.05, 0.07, 0.10, 0.13, 0.17, 0.20, 0.25, 0.33, 0.50, 1.00}

3. **20 frames** at each exposure step. 3 fields × 12 steps × 20 = 720 frames per night.
4. Record for every frame: exposure, field declination, timestamp, and **tracking state = off**.
5. **Position angle control.** The trail direction is along the hour-angle axis, which for an alt-az tabletop mount rotates relative to the sensor as the target moves. Shoot the same field early and late in the evening to get two different position angles — this prevents the model from learning "elongation means horizontal".
6. **Negatives.** Add either (a) $t \le 0.02$ s frames, where the trail is under half a pixel, or (b) frames from the tracked loaner rig. At least 1,000 negative frames.
7. **Confounders (one session, deliberately):**
   - Frames using the *edge* of the field, where an f/3.95 Newtonian's coma elongates stars radially — a different position angle signature from tracking error.
   - A windy night, which produces incoherent smear rather than a linear trail.

- A deliberate mid-exposure nudge (tap the tube), giving a double image rather than a streak.

Recorded. Frames + exposure + declination + tracking state + wind estimate + which confounder class, if any.

Abort criteria. Cloud; star count < 10 per frame.

Acceptance. Measured ellipticity versus predicted ℓ_{px} follows the expected curve within scatter. **If it does not, the plate scale from BP-4 is wrong** — and finding that out here is much cheaper than finding it out during training.

RB-7 — Sky background and lunation ladder

Purpose. Populate the sky-brightness covariate that both the exposure bandit and the filter recommender consume, and validate the classical background estimator against real gradients.

Preconditions. None beyond a body and tripod. Runs alongside RB-5 on the same frames if the settings are held fixed.

1. Same fixed field, same exposure, same ISO, **once per hour, all night**.
2. Repeat across a full lunation: at least one night near new moon, one at first quarter, one near full, one at last quarter. Four nights spread over four weeks.
3. Include one session pointed low toward the light dome and one high toward zenith on the same night — the gradient between them is the light-pollution signal.
4. Record moon altitude, illuminated fraction, and moon–field angular separation (all computable from time and site, but record the site coordinates).

Acceptance. Background ADU varies monotonically with moon illumination at fixed altitude. If it does not, something is auto-adjusting — check that exposure, ISO and white balance are truly locked.

RB-8 — Exposure bandit logging (R5, rental)

Purpose. Build a logged-bandit dataset that supports *offline* policy evaluation, so that a policy can be validated without spending rental nights on live exploration.

Preconditions. Tracked mount; a body that tethers with bulb (T3i/T6S are the safest bets after BP-1); D6 ideally resolved so airmass and moon separation come from a solved pointing.

1. **Phase 1 — uniform logging.** For the first 200 subs, select the exposure arm **uniformly at random** from $\{10, 30, 60, 90, 120, 180, 300\}$ s. Record the arm *and* its selection probability $1/7$.
2. **Phase 2 — ε-greedy.** Thereafter use $\varepsilon = 0.2$: with probability 0.2 choose uniformly, otherwise choose the current best. **Record the realised probability for the chosen arm on every decision** — this is the propensity, and without it no unbiased off-policy estimate exists.
3. **Record the 17-dim context** at decision time, before the exposure: sky background, moon illumination, $\cos\theta_{\text{moon}}$, filter one-hot, target-type one-hot, seeing (star FWHM), airmass, guiding RMS. If there is no guider, record guiding RMS as **absent**, not zero.
4. **Record all five reward components separately** — sub accepted, SNR in range, sub rejected, saturation $> 5\%$, trailing — in `frame_measurements.payload_json`. Never store only the scalar sum.
5. Vary target type across sessions — galaxy, nebula, open cluster, globular — since target type is four of the seventeen context dimensions and a single-target campaign leaves three of them constant.

Acceptance. Every logged tuple carries a propensity. Every arm has been selected at least 20 times. At least three distinct moon phases and two target types are represented.

Abort criteria. If the propensity is not being logged, **stop and fix it before continuing** — data without it cannot be used for anything except online learning, and the entire reason for the campaign is to avoid learning online on rented time.

RB-9 — Filter sequence (descoped)

Documented for completeness; not scheduled. If a loaner rig arrives with a filter wheel, log every switch through `filter_switch_events` (schema already present, migration 003) with: time, from-filter, to-filter, reason, sky conditions at the time, and the realised SNR of the subs on each side. That costs nothing and preserves the option. Do not purchase hardware for FOR — see §7.8.

Appendix C — Observation log schema

C.1 Session header (once per night)

Field	Type	Note
`session_id`	uuid	generated
`night_id`	date	local date of *sunset*, so one night is one ID across midnight
`rig_id`	string	R1–R5, matching §3.5
`site_lat`, `site_lon`, `site_elev`	float	needed for airmass and moon geometry
`telescope`, `camera`, `barlow`, `focuser`	string	the exact optical train
`plate_scale_arcsec_px`	float	from BP-4, per configuration
`saturation_adu`	int	from BP-2, per body and ISO
`observer`	string	
`start_time`, `end_time`	ISO 8601 with offset	

C.2 Environment series (every 15 minutes)

`timestamp`, `ambient_temp_c`, `humidity_pct`, `dew_point_c`, `wind_beaufort`, `sky_class` (0–4 per RB-5), `transparency_naked_eye_limiting_mag`, `seeing_impression` (1–5, subjective, but recorded consistently), `notes`.

> Every one of these may be **absent**. Absent is a value; write `null`, not a guess. `frames.temperature_state` already models exactly this distinction and the log must not undermine it.

C.3 Run record (per focus run or per campaign block)

`run_id`, `run_type` (`focus_sweep` | `trailing_ladder` | `cloud_block` | `drift` | `bandit` | `calibration`), `target_name`, `target_ra`, `target_dec`, `target_alt`, `target_az`, `start_position`, `step_size`, `n_positions`, `direction`, `tracking_on`, `exposure_s`, `iso_or_gain`, `filter`, `status` (`completed` | `aborted`), `abort_reason`.

C.4 Disturbance record (as they happen)

`timestamp`, `kind` (`bump` | `refocus` | `cloud` | `headlights` | `dew` | `battery` | `wind_gust` | `other`), `affected_run_id`, `description`.

This is the block that gets skipped and should not be. One unlogged bump in the middle of a sweep produces a bimodal V-curve that the fitter will resolve to the wrong minimum, and there is no way to detect it from the pixels alone.

C.5 Machine-readable form

The sidecar importer (§G2/G4) consumes one JSON file per session:

```
{
  "session": { "session_id": "...", "night_id": "2026-09-14",
               "rig_id": "R2", "site_lat": 36.07, "site_lon": -79.81,
               "telescope": "Zhumell Z114 + 2x Barlow",
               "camera": "SVBony SV105",
               "plate_scale_arcsec_px": 0.71,
               "saturation_adu": 255 },
  "environment": [
    { "t": "2026-09-14T21:15:00-04:00", "ambient_temp_c": 21.4,
      "humidity_pct": 68, "wind_beaufort": 1, "sky_class": 0,
      "seeing_impression": 3 },
    { "t": "2026-09-14T21:30:00-04:00", "ambient_temp_c": 20.9,
      "humidity_pct": 71, "wind_beaufort": 1, "sky_class": 1,
      "seeing_impression": 3 }
  ],
  "runs": [
    { "run_id": "...", "run_type": "focus_sweep", "target_name": "M39",
      "start_position": 1240, "step_size": 14, "n_positions": 21,
      "direction": "outward", "tracking_on": false,
      "exposure_s": 0.15, "iso_or_gain": 300, "status": "completed",
      "positions": [
        { "index": 0, "reading_um": 1240, "t": "...", "files": ["IMG_0001.CR2"] }
      ]
    }
  ],
  "disturbances": [
    { "t": "2026-09-14T23:02:00-04:00", "kind": "headlights",
      "affected_run_id": "...", "description": "car in driveway, 2 frames" }
  ]
}
```

Fill the card on paper in the field and transcribe to JSON indoors the same night. Do not plan to remember it in the morning — you will not, and the transcription step is also a useful review that catches missing entries while the night is still recoverable.

Appendix D — Post-session ingest checklist

1. **Do not format the card.** Copy files off it; keep the originals until the checksums verify.
2. Transcribe the paper card to the JSON schema of §C.5 **the same night**.
3. Copy frames to the capture root, preserving the camera's filenames.
4. Run the sidecar importer for anything the daemon did not capture itself. Confirm it stamps `driver_name = "offline_import"`.
5. Start the daemon and let `reconcile()` run. Read its log — it reports committed rows with missing files and staged files with no row.
6. Verify checksums against `frame_assets.checksum_sha256`. A mismatch means a bad copy; recopy from the card, which is why step 1 exists.
7. Count what you got, against what you attempted:

```
sqlite3 achernar.db "
  SELECT capture_type, capture_status, COUNT(*)
  FROM frames WHERE date(captured_at)='2026-09-14'
  GROUP BY 1,2;"
sqlite3 achernar.db "
  SELECT status, COUNT(*) FROM capture_attempts
  WHERE date(started_at)='2026-09-14' GROUP BY 1;"
```

8. Dry-run the dataset builder and **check the group count, not the row count**:

```
python -m ml.data_pipeline.build_focus_quality_dataset \
  --db achernar.db --output /tmp/check.npz
python3 -c "import numpy as np; d=np.load('/tmp/check.npz',allow_pickle=True); \
  print('rows', d['crops'].shape[0], 'groups', len(set(d['run_key'].tolist())))"
```

9. Confirm the saturation gate actually ran on this night's frames. If it reports *not performed*, D1 or D2 is still open and the night's focus labels are suspect.
10. Back up: capture root + `achernar.db` + the session JSON, to a second physical drive. Only then reuse the card.
11. Update the running tally — runs per rig, frames per COC class, nights per campaign — against the budgets in Part III, and pick the next night's target accordingly.

Appendix E — Standing decisions

Decision	Rationale
Collect COC labels as 5-class	Collapsible to the spec's binary head; the reverse is impossible, and the sky is not re-observable (D7).
2× Barlow is mandatory on the Z114	Native focal length is undersampled at best focus for every available camera (§3.4).
Descope FOR (Model 8)	No wheel, no filters, `hardware/filter_wheel.rs` is `todo!()`, and its features depend on models that do not exist yet (§7.8).
TME edge by distillation, not contrastive	AstroM3's published weights already exist; only the image adapter is untrained, and 10^5 contrastive objects are out of reach (§7.9).
FOP data by passive harvesting	20 rows per run cannot reach 10^4 from dedicated sweeps alone (§7.2).
Ambient temperature logged by hand for v1	`hardware/sensors.rs` is `todo!()`; manual logging is well within the Kalman filter's assumed $\sigma_z = 3$ steps (§7.3).
Dial indicator instead of a motorised focuser for v0/v1	\$15 buys 10 µm absolute position, which makes manual runs FOP-eligible (RB-2).
Rental window spent last	It is the scarce resource; field procedures should be debugged on free gear first (§8.2).

Open questions for you to resolve, in priority order.

1. Does the loan programme's rig include an electronic focuser, and on what protocol? This decides whether \$200–300 needs spending and whether T2 and T3 merge.
2. Does `cargo build` pass on the Mac (BP-0)? Everything downstream is contingent.
3. Can libuvc open the SV105 on macOS (BP-1)? If not, R1 and R2 need a Linux host.
4. Which of the five bodies actually support remote capture (BP-1)?
5. Does any camera reach prime focus on the Z114 (BP-3)?